The Privacy–Utility Trade-off, Worked Through

One use case and two routes, a private model or a private synthetic table, from the business question to a signed-off decision

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A bank's outside agency wants either a targeting model or the customer table itself, and the bank's policy requires differential privacy (DP) for anything that leaves. Someone other than the engineer has to choose how much privacy to buy and what to release. This note works through that choice on a public dataset, from a business count and a sentence a privacy officer can sign to one decision between a private model and a private synthetic table.

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How strict should a differential privacy (DP) guarantee be, what form should a private release

take, and who decides? This note answers these questions for a bank's credit-card campaign and ends with a decision signed by the people responsible for its risks and revenue. You need basic probability, including odds, averages and percentiles. Table 1 lists each section's question, and Table 2 defines the notation.

Table 1: The question each section answers.

Section	The question it answers
Section 1	What does the bank want, what can it release, and which public data stands in for its records?
Section 2	Which counts does the business care about, and what must a release beat?
Section 3	What does ε mean to a privacy officer, and which value does the policy allow?
Section 4	Which choices must be fixed before anyone touches the records, so that they leak nothing?
Section 5	How does the bank train a private model?
Section 5.2	Which model technique is best at each privacy level, and what does privacy cost in customers?
Section 5.3	How does that cost change with the number of customers?
Section 6.1	How does a generator of synthetic data account for its privacy?
Section 6.2	Which three generators are compared, and how do they differ?
Section 6.3	Which generator is best for the plan and for the campaign?
Section 6.4	How does the synthetic table's cost change with the number of customers?
Section 7	Which route and which option does the policy pick, and who signs it?
Section 8	How do the steps carry over to another business?
Section 9	Where do the data, the methods and the arguments come from?

Table 2: The symbols the note uses. Each is also explained where it first appears.

Symbol	Meaning
ε	the privacy parameter of DP: a smaller value means stronger privacy
e^{ε}	the most that seeing the release can multiply an attacker's odds by
δ	the small probability that the guarantee fails outright

Symbol	Meaning
ρ	the privacy budget of a generator, in zero-concentrated DP
σ	the standard deviation of the noise that a generator adds to each count
V	what the bank gains when it reaches one eligible customer instead of an ineligible one
k	a difference in the number of eligible customers reached

1 The use case

A retail bank is about to launch a premium credit card for customers who earn more than $50,000 a year, and an outside agency will run the campaign. The bank does not record income, but it holds other facts about each customer, such as age, education, occupation, hours worked and marital status. So the campaign needs a model that predicts from those facts who is likely to be eligible. The agency would also like to plan the campaign, which means to count the eligible customers in each segment it might target.

The bank's privacy policy does not let customer records leave its secure environment. Anything that does leave must be trained or generated with DP, so that nobody can use it to learn much about any one customer. The bank can meet the policy in two ways. On the *model route*, it trains the targeting model itself with DP and hands over the model alone. On the *synthetic-data route*, it hands over a *synthetic table*, which has the same columns as the records but rows that a DP procedure generates.

The agency can count, plot and train on a synthetic table as it likes, because anything computed from a DP release has the same guarantee. The policy does not say how strict the guarantee must be, nor which route to take. Those two choices are the subject of this note.

The bank's records are private, so we use a public dataset like them, called Adult. Adult holds 48,842 people from the 1994 US Current Population Survey. Each has these attributes and a label for income above $50,000 (UCI Machine Learning Repository). Its training file of 32,561 people plays the bank's customer records, which the model or the synthetic table comes from. Its test file of 16,281 people plays the customers of the next campaign.

Two scripts compute every number below, one for each route, and each step shows their lines. code/notes/privacy_utility_tradeoff.py trains the private models in a few minutes on all of a machine's processor cores. code/notes/privacy_utility_tradeoff_synthetic.py runs the generators. It runs four processes at a time, because each loads JAX and PyTorch, and `PRIVACY_WORKERS` changes that number:

```
uv run python code/notes/privacy_utility_tradeoff.py
uv run python code/notes/privacy_utility_tradeoff_synthetic.py
```

2 Step 1: state the business goal as counts

Engineers usually measure a model by its accuracy or area under the curve (AUC), and a synthetic table by how closely its statistics match the records. Neither measure directly tells the marketing team how many eligible customers an offer campaign will reach. This step fixes the number of offers and counts the eligible customers among recipients.

The bank can choose the number of offers from public data. The Census Bureau publishes the share of people who earn more than $50,000, about 24% in Adult. If the bank sends as many offers as it expects eligible customers, a perfect model wastes none. For the 16,281 campaign customers this rule gives 3,846 offers, the number of eligible customers among them, when the published share holds for the bank.

The bank sends the 3,846 offers to the customers with the highest scores. The business metric is the number of eligible customers among them, and every other offer is wasted. The script computes it with a short function:

```
def reached(scores: Array, y: npt.NDArray[np.int64], offers: int) -> int:
    """Eligible customers among the `offers` highest-scored; ties are broken by
     ↪  position, deterministically."""
    top = np.argsort(-scores, kind="stable")[:offers]
    return int(y[top].sum())
```

Use random selection as the baseline. If the bank sent the offers to customers picked at random, each would reach an eligible customer with probability 3,846 / 16,281, about 24%. The expected number reached would be

$$3{,}846 \times \frac{3{,}846}{16{,}281} \approx 909. \tag{1}$$

Any release, a private model or a synthetic table, has to beat 909 to be worth having. The script sets up the campaign in the same way:

```
df = practitioner_datasets.adult()
train, test = df.head(TRAIN_ROWS), df.slice(TRAIN_ROWS)
data = (features(train), labels(train), features(test), labels(test))
eligible = int(data[3].sum())
# Random offers reach an eligible customer at the base rate, eligible / campaign
 ↪  customers.
baseline = eligible * eligible / len(data[3])
```

A synthetic table also serves the agency's plan, and that gives a second count. The agency divides the customers into segments by age band and occupation, six age bands times fifteen occupations, which makes 90 segments. The records hold 7,841 eligible customers, and the plan says how many of them each segment holds. A synthetic table gives a different spread, so its error is the number of eligible customers it puts in the wrong segment. We count that as half the sum, over the segments, of the absolute difference between the synthetic count and the real count. When the two tables hold the same number of eligible customers, it is the number of customers that must move between segments to turn one plan into the other.

The second script computes this count, and also the error on the two-way tables that Section 6.3 uses for the technical view:

```
def workload_error(real: Dataset, synth: Dataset) -> float:
    """Average over the two-way tables of the total variation distance between real and
     ↪  synthetic proportions."""
    errors: list[float] = []
    for pair in pairs(real.domain):
        x = np.asarray(real.project(pair).datavector(), dtype=np.float64)
        y = np.asarray(synth.project(pair).datavector(), dtype=np.float64)
        errors.append(0.5 * float(np.abs(x / x.sum() - y / y.sum()).sum()))
    return float(np.mean(errors))


def misplaced(real: Dataset, synth: Dataset) -> float:
    """Eligible customers the synthetic plan puts in the wrong segment: half the L1
     ↪  distance between the plans."""
    return 0.5 * float(np.abs(plan(real) - plan(synth)).sum())
```

The counts also give the money. Write V for what the bank gains when it reaches one eligible customer instead of an ineligible one. It is the chance that an eligible customer takes up the card times the card's margin, and the finance team estimates it. Every comparison below is a difference of k eligible customers reached, worth kV to the bank. So the note never needs a value for V, because the decision in Section 7 only compares kV with what the policy protects.

A misplaced customer is harder to price, because it costs the bank only when the agency sizes a segment wrongly. So we report the plan's error as a count and as a share of the 7,841, and the agency says what share it accepts.

3 Step 2: turn the privacy setting into a sentence

DP is set by one number, ε, and a smaller ε means stronger privacy. A privacy officer cannot sign a bare number, so this step translates ε into an attacker's question. Suppose an attacker wants to know whether one person, say Alice, is a customer of the bank, and starts with some belief about it. DP guarantees that seeing the release, the model or the table, can multiply the attacker's odds by at most e^{ε}.

The simplest case is an attacker who starts at even odds, a belief of 50%. The odds are then 1 to 1, and after the attacker sees the release they are at most e^{ε} to 1. Odds of e^{ε} to 1 are a probability of e^{ε} out of $1 + e^{\varepsilon}$, so the attacker's belief can rise to at most

$$\text{belief ceiling} = \frac{e^{\varepsilon}}{1 + e^{\varepsilon}}. \tag{2}$$

At $\varepsilon = 1$ the ceiling is $e/(1 + e)$, about 73%. At $\varepsilon = 0.1$ it is about 52.5%, so the release teaches the attacker almost nothing. At $\varepsilon = 3$ it is about 95%, a guarantee that still holds but is weak.

The privacy officer signs this sentence, and in our use case they choose a ceiling of 75% for both routes. In other words, an attacker who thinks it is a coin flip whether Alice is a customer must

not end up more than 75% sure. To turn the sentence into a number, multiply out the fraction to get a bound on e^{ε}, then take the logarithm:

$$\frac{e^{\varepsilon}}{1+e^{\varepsilon}} \le 0.75 \quad \Longleftrightarrow \quad e^{\varepsilon} \le 3 \quad \Longleftrightarrow \quad \varepsilon \le \ln 3 \approx 1.10. \tag{3}$$

The scripts convert between ε and the belief ceiling. `belief_ceiling` labels the top axis of each figure, and `epsilon_for_belief(0.75)` returns the policy's limit of 1.10:

```
def belief_ceiling(eps: float, prior: float = 0.5) -> float:
    """Highest posterior that epsilon-DP allows an attacker who starts at `prior` that
     ↪  one customer is in the data."""
    odds = math.exp(eps) * prior / (1 - prior)
    return odds / (1 + odds)


def epsilon_for_belief(belief: float, prior: float = 0.5) -> float:
    """The largest epsilon whose belief ceiling, from `prior`, is `belief`."""
    return math.log((belief / (1 - belief)) / (prior / (1 - prior)))
```

The officer fixes the ceiling before anyone sees the results. Choosing it afterwards would bias the policy toward a release that performs well. Both routes also use δ, the small probability that the guarantee fails outright. A common rule sets δ below one over the number of people, so that the guarantee is not expected to fail for anyone. For 32,561 customers this gives $\delta = 10^{-5}$, which we use throughout.

4 Step 3: fix everything that must not be tuned on private data

A DP guarantee covers data access during training or generation, but not choices made after examining the records outside that private computation. For example, trying twenty settings on the customer records and keeping the best one would itself leak information. Every choice outside the private run must therefore use public information.

For the model route, the preprocessing comes from the survey's public codebook, not from the data. It scales ages within 17 to 90 and years of education within 1 to 16. It scales weekly hours within 1 to 99, and capital gains and losses on a log scale up to the survey's published cap of 99,999. Each category, such as an occupation, becomes a 0-or-1 column from the codebook's list. Every customer's row therefore lies in a known range. DP needs this range to bound how much one customer can change the training.

The preprocessing function reads only these public ranges and lists, never a statistic of the data:

```
def features(df: pl.DataFrame) -> Array:
    """Map each row to [0, 1]^d using only the codebook; a missing category is a row of
     ↪  zeros."""
    cols: list[Array] = []
    for name, (lo, hi) in LINEAR_BOUNDS.items():
        x = df[name].cast(pl.Float64).to_numpy()
        cols.append(np.clip((x - lo) / (hi - lo), 0.0, 1.0))
```

```
    for name in LOG_COLUMNS:
        x = df[name].cast(pl.Float64).to_numpy()
        cols.append(np.clip(np.log1p(np.clip(x, 0, TOP_CODE)) / math.log1p(TOP_CODE),
↪   0.0, 1.0))
    for name, values in CATEGORIES.items():
        column = df[name].to_list()
        for value in values:
            cols.append(np.array([1.0 if v == value else 0.0 for v in column]))
    return np.column_stack(cols)
```

The synthetic-data route needs a table of discrete attributes, so the numbers go into bands and the categories again come from the codebook. Age goes into six bands, from 17 to 24 up to 65 and over, and hours worked into five. Capital gains and losses, which are zero for most people, go into four bands each. The eight categorical attributes keep the codebook's lists, with a "missing" category where the survey allows a blank. With the income label that makes 13 attributes, and the survey weight and the numeric copy of education are dropped.

The product of the 13 attributes' sizes is about 37 billion, so each row is one cell of a table with 37 billion cells. That is why no generator can simply add noise to the count in every cell. The encoding function reads only the bands and the lists:

```
def encode(df: pl.DataFrame) -> Dataset:
    """Map each row to one integer per attribute: its band, or its category's position
    ↪  in the codebook's list."""
    columns: dict[str, Ints] = {}
    for name, limits in BANDS.items():
        x = df[name].cast(pl.Int64).to_numpy()
        columns[name] = np.searchsorted(np.array(limits), x,
↪   side="left").astype(np.int64)
    for name, values in CATEGORIES.items():
        index = {v: i for i, v in enumerate(values)}
        missing = index.get("missing", -1)
        columns[name] = np.array([index[v] if v is not None else missing for v in
↪   df[name].to_list()], dtype=np.int64)
        assert columns[name].min() >= 0, f"a value of {name} is not in the codebook"
    return Dataset(columns, domain())
```

Each route's techniques and their settings are also fixed in advance, and the next two steps describe them.

5 Step 4: the model route

5.1 Train a private model

The model route compares two techniques. The first is private stochastic gradient descent (SGD), known as **DP-SGD**, the standard way to train a model with DP. It trains a logistic regression, as a bank would without DP, on small random batches. At each step it caps how much each

customer can move the model, which is called *clipping*, and it adds random noise to the update (Opacus).

Opacus's privacy accountant sets the noise so that the whole run meets the target ε at $\delta = 10^{-5}$. From ε, δ, the share of customers in each batch and the number of passes, it returns the noise multiplier, the size of the noise relative to the clipping norm:

```
def noise_multiplier(eps: float, n: int) -> float:
    """The noise multiplier that makes DP-SGD with these settings (eps, DELTA)-DP, by
     ↪  Opacus's PRV accountant."""
    return float(
        get_noise_multiplier(
            target_epsilon=eps, target_delta=DELTA, sample_rate=BATCH / n,
 ↪  epochs=EPOCHS, accountant="prv"
        )
    )
```

Opacus then wraps the model, the optimizer and the data loader, so that every step draws a random batch, clips each customer's contribution and adds that noise. The rest of the loop is the same as without DP:

```
optimizer = torch.optim.SGD(layer.parameters(), lr=LR, momentum=MOMENTUM)
engine: PrivacyEngine | None = None
model: nn.Module = layer
if sigma is not None:
    engine = PrivacyEngine(accountant="prv")
    model, optimizer, loader = engine.make_private(  # Poisson sampling, per-example
 ↪  clipping, Gaussian noise
        module=layer, optimizer=optimizer, data_loader=loader, noise_multiplier=sigma,
 ↪  max_grad_norm=CLIP
    )
```

DP-SGD's settings were fixed in advance, with batches of about 1,000 customers, 10 passes over the data, a clipping norm of 1 and a learning rate of 0.5. The second technique is a **private naive Bayes** classifier from IBM's diffprivlib, which adds noise to simple averages per class (diffprivlib). It gives the stronger pure ε guarantee, with no δ. A pure guarantee also meets the $(\varepsilon, 10^{-5})$ standard, so both techniques sit on the same axis.

The experiment trains each technique at ten values of ε from 0.01 to 10. DP training is random, and one run can be lucky or unlucky, so each setting runs with 20 random seeds. Each model is also trained without DP, with the same settings, so that the figures show the cost of privacy directly. The script spreads these runs over the processor cores:

```
jobs: list[tuple[str, float | None, int, int]] = []
for technique in COLOURS:
    jobs += [(technique, eps, TRAIN_ROWS, seed) for eps in [None, *EPSILONS] for seed
 ↪  in range(seeds)]
for n in SIZES[:-1]:
    jobs += [("DP-SGD", eps, n, seed) for eps in [None, *EPSILONS] for seed in
 ↪  range(size_seeds)]
```

```
with ProcessPoolExecutor(max_workers=WORKERS, initializer=_init, initargs=(data,)) as
↪   pool:
    runs = list(pool.map(evaluate, jobs, chunksize=4))
```

5.2 Read the results, technical first

Figure 1 shows the results in the engineer's terms. Its vertical axis is the AUC, the probability that the model scores a randomly chosen eligible customer above a randomly chosen ineligible one, so 0.5 is guessing and 1 is perfect. The bottom axis is ε, and the top axis gives the attacker's belief ceiling from Section 3. The shaded bands run from the 10th to the 90th percentile over the seeds.

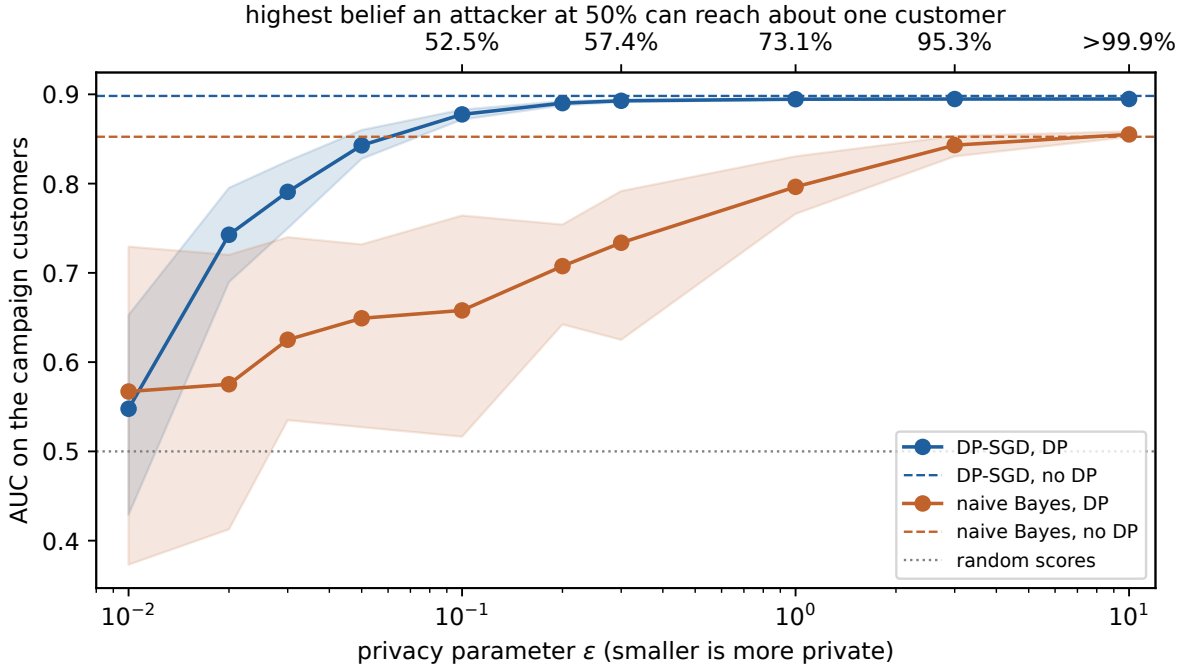


Figure 1: Model quality against the privacy setting. Each line is a DP technique, averaged over 20 seeds, with its 10th–90th percentile band; each dashed line is the same model trained without DP.

DP-SGD reaches higher AUC than naive Bayes at every privacy level except $\varepsilon = 0.01$, where their results are nearly tied. Naive Bayes therefore adds little at the other levels. DP-SGD's AUC changes little over most of the range; above about $\varepsilon = 0.3$, it nearly matches the model trained without DP. Its AUC declines below $\varepsilon \approx 0.1$, where the attacker's belief ceiling is already within a few percentage points of 50%.

Figure 2 shows them in the bank's terms. Its vertical axis is the number of eligible customers reached by the 3,846 offers. The upper dashed line is the best model without DP, the dotted line is no model at all, and the shaded region on the right is what the privacy policy forbids.

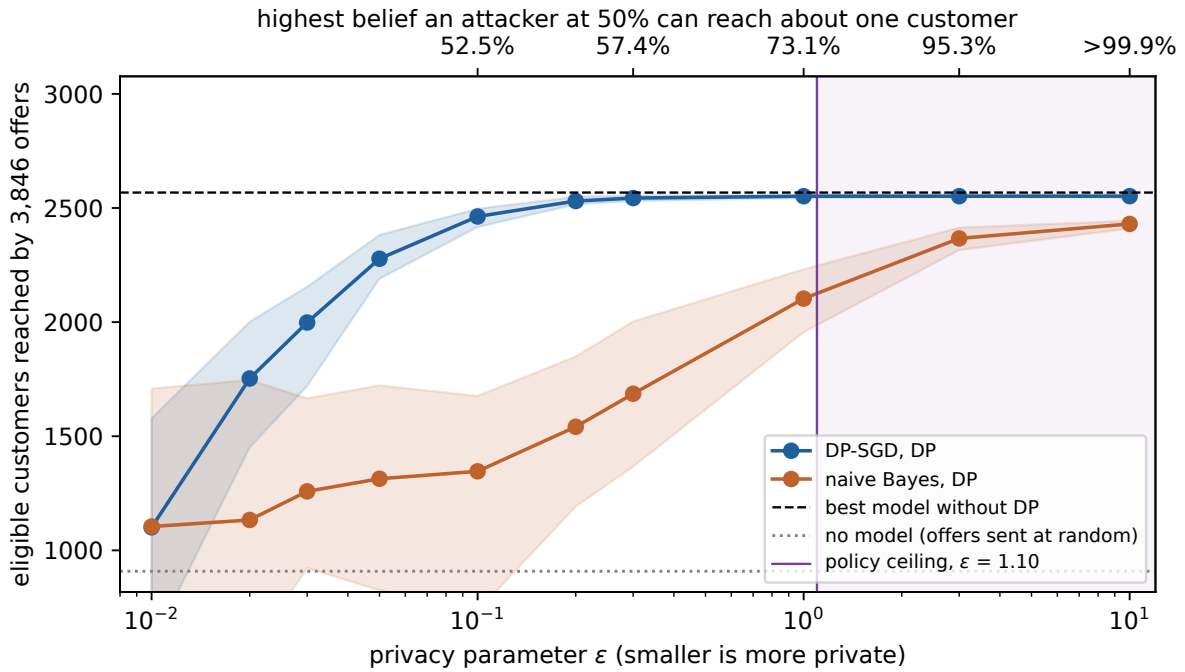


Figure 2: Eligible customers reached by the campaign against the privacy setting. The shaded region is ruled out by the policy ceiling.

The gap between a DP curve and the upper dashed line is the cost of privacy, counted in customers. The gap between the curve and the dotted line is what the model is worth compared with random offers. Table 3 lists every private configuration. A configuration is on the *frontier* when no other configuration is at least as private and reaches more customers, so only the frontier is worth considering. Here the frontier is DP-SGD at every level from $\varepsilon = 0.02$ up, with naive Bayes only at $\varepsilon = 0.01$.

The script finds the frontier with a short filter:

```
def non_dominated(points: list[tuple[float, float]]) -> list[int]:
    """Indices of the (privacy loss, utility) points that no other point dominates."""
    order = sorted(range(len(points)), key=lambda i: (points[i][0], -points[i][1]))
    frontier: list[int] = []
    best = float("-inf")
    for i in order:
        if points[i][1] > best:
            frontier.append(i)
            best = points[i][1]
    return frontier
```

Table 3: Every private configuration, averaged over the seeds.

Technique	ε	Attacker's belief ceiling	AUC	Eligible reached	On the frontier
DP-SGD	0.01	50.2%	0.548	1,101 of 3,846 (29%)	
naive Bayes	0.01	50.2%	0.567	1,104 of 3,846 (29%)	yes
DP-SGD	0.02	50.5%	0.743	1,753 of 3,846 (46%)	yes
naive Bayes	0.02	50.5%	0.575	1,133 of 3,846 (29%)	
DP-SGD	0.03	50.7%	0.791	1,998 of 3,846 (52%)	yes
naive Bayes	0.03	50.7%	0.625	1,259 of 3,846 (33%)	
DP-SGD	0.05	51.2%	0.843	2,278 of 3,846 (59%)	yes
naive Bayes	0.05	51.2%	0.649	1,314 of 3,846 (34%)	
DP-SGD	0.1	52.5%	0.878	2,463 of 3,846 (64%)	yes
naive Bayes	0.1	52.5%	0.658	1,346 of 3,846 (35%)	
DP-SGD	0.2	55.0%	0.890	2,531 of 3,846 (66%)	yes
naive Bayes	0.2	55.0%	0.708	1,542 of 3,846 (40%)	
DP-SGD	0.3	57.4%	0.893	2,544 of 3,846 (66%)	yes
naive Bayes	0.3	57.4%	0.734	1,686 of 3,846 (44%)	
DP-SGD	1	73.1%	0.895	2,552 of 3,846 (66%)	yes
naive Bayes	1	73.1%	0.796	2,103 of 3,846 (55%)	
DP-SGD	3	95.3%	0.895	2,552 of 3,846 (66%)	yes
naive Bayes	3	95.3%	0.843	2,367 of 3,846 (62%)	
DP-SGD	10	100.0%	0.895	2,552 of 3,846 (66%)	yes
naive Bayes	10	100.0%	0.855	2,430 of 3,846 (63%)	

5.3 Check how much the data size matters

The flat curve in Figure 2 depends on the bank having many customers. DP hides each person with noise, and the more customers there are, the less that noise matters next to their combined signal. So a bank with fewer customers, or a campaign aimed at one region, pays more for the same guarantee. To measure how much, the script trains DP-SGD again on the first 2,000 and the first 8,000 customers, and Figure 3 compares the three sizes.

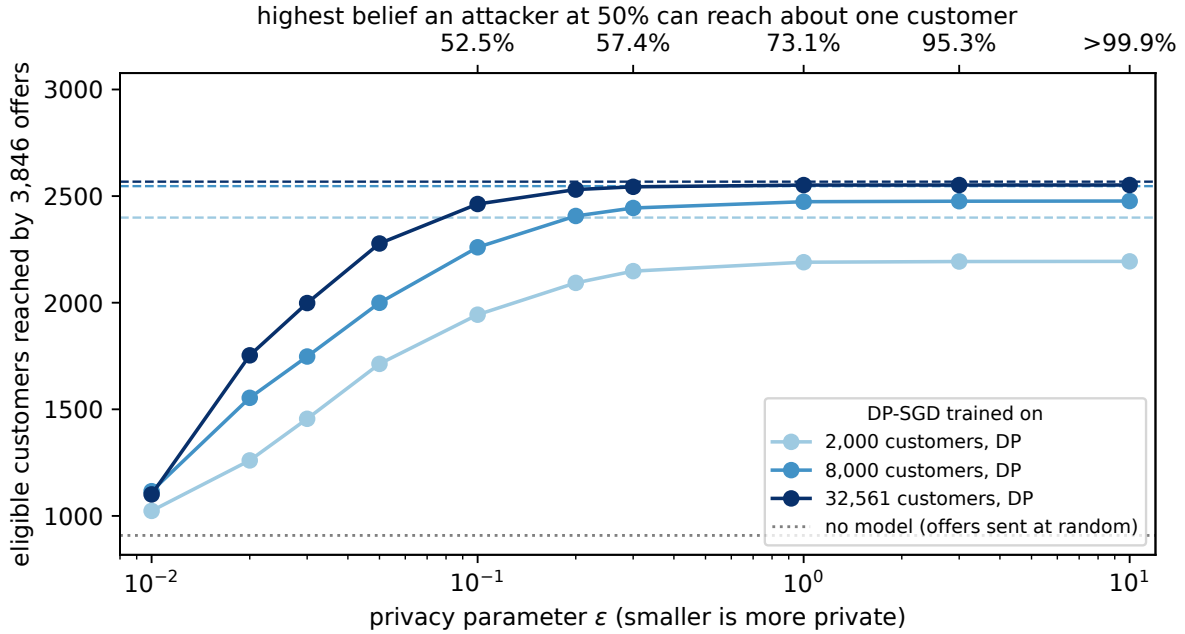


Figure 3: Eligible customers reached by DP-SGD trained on 2,000, 8,000 and 32,561 customers. Each dashed line is the same size without DP.

With fewer customers, performance falls at larger ε, and the gap from the model without DP is wider at every setting, as Table 4 shows. For the smallest dataset, a gap remains even at $\varepsilon = 10$, where noise is negligible. The clipping settings were chosen for the full dataset, so they contribute to this gap. A bank with fewer customers should first tune them on public or synthetic data. This comparison shows whether the guarantee is affordable for the dataset and whether more data would make a stricter policy feasible.

Table 4: Eligible customers missed because of privacy, by the number of customers the model is trained on.

Customers trained on	Reached without DP	Missed at $\varepsilon = 0.1$	Missed at $\varepsilon = 0.3$	Missed at $\varepsilon = 1$
2,000	2,399	455	251	209
8,000	2,547	287	102	73
32,561	2,568	105	24	16

6 Step 5: the synthetic-data route

6.1 Give the generators a privacy budget

The generators add Gaussian noise repeatedly, and ε does not compose those measurements tightly. Instead, each measurement is accounted for in *zero-concentrated differential privacy* (zCDP). Its budget ρ has a single parameter, and measurement costs add. Let σ be the standard deviation of noise added to a count that one customer can change by one. The measurement costs $\rho = 1/(2\sigma^2)$, and the costs of multiple measurements sum.

A total budget ρ converts to (ε, δ) by a formula that gives the smallest δ for each ε. The script runs that conversion backwards, by bisection, and finds the largest ρ that the policy's ε and δ allow:

```python
def zcdp_delta(rho: float, eps: float) -> float:
    """The smallest delta for which rho-zCDP implies (eps, delta)-DP (Canonne, Kamath &
     ↪ Steinke, 2020, Prop. 7)."""
    lo, hi = 1.01, (eps + 1) / (2 * rho) + 2  # the best order alpha lies in this range
    alpha = (lo + hi) / 2
    for _ in range(100):
        alpha = (lo + hi) / 2
        if (2 * alpha - 1) * rho - eps + math.log1p(-1 / alpha) < 0:
            lo = alpha
        else:
            hi = alpha
    return min(1.0, math.exp((alpha - 1) * (alpha * rho - eps) + alpha * math.log1p(-1
     ↪ / alpha)) / (alpha - 1))


def rho_for_epsilon(eps: float, delta: float = DELTA) -> float:
    """The largest zCDP budget rho that (eps, delta)-DP allows, found by bisection."""
    lo, hi = 0.0, eps + 1
    for _ in range(100):
        rho = (lo + hi) / 2
        lo, hi = (rho, hi) if zcdp_delta(rho, eps) <= delta else (lo, rho)
    return lo
```

Table 5 lists the budgets for three values of ε at $\delta = 10^{-5}$. Every generator receives ρ and spends it, so each is $(\varepsilon, 10^{-5})$-DP by construction, on the same axis as the model route.

Table 5: The zero-concentrated budget that the policy's $(\varepsilon, 10^{-5})$ allows.

ε	Attacker's belief ceiling	Budget ρ
0.1	52.5%	0.00043
1	73.1%	0.031
3	95.3%	0.22

6.2 Fix three generators

All three generators measure *marginal tables*. A one-way table counts the customers with each value of one attribute. A two-way table counts them for each pair of values of two attributes, such as each combination of age band and occupation. The 13 attributes give 78 two-way tables, the largest of which, country by education, has 672 cells. One customer sits in exactly one cell of each table, so adding or removing a customer changes one count by one. So noise of standard deviation σ on every cell costs $\rho = 1/(2\sigma^2)$, whatever the table's number of cells.

The noisy tables then go to Private-PGM (McKenna, Sheldon & Miklau, 2019). It finds the probability distribution over whole rows whose marginals best match the noisy tables, and it samples synthetic rows from it. Everything after the measurements is computation on noisy numbers, so it costs no privacy.

The script measures a table with one function and chooses among candidate tables with another. The second is the *exponential mechanism*. It draws a candidate with probability proportional to $e^{\varepsilon \cdot \text{score}/2}$, so a candidate with a higher score is more likely but never certain, and that keeps the choice private when the scores depend on the data:

```
def measure(data: Dataset, clique: Clique, sigma: float, rng: np.random.Generator) ->
 ↪  LinearMeasurement:
    """The table of counts over `clique`, with Gaussian noise of scale sigma on every
     ↪  cell: rho = 1 / (2 sigma^2)."""
    counts = np.asarray(data.project(clique).datavector(), dtype=np.float64)
    return LinearMeasurement(counts + rng.normal(0.0, sigma, counts.size), clique,
     ↪  stddev=sigma)


def exponential_mechanism(scores: dict[Clique, float], eps: float, rng:
 ↪  np.random.Generator) -> Clique:
    """One candidate, drawn with probability proportional to exp(eps * score / 2); each
     ↪  score has sensitivity 1."""
    cliques = list(scores)
    logits = np.array([eps * scores[c] / 2 for c in cliques])
    weights = np.exp(logits - logits.max())
    return cliques[rng.choice(len(cliques), p=weights / weights.sum())]
```

The generators differ only in which tables they measure. A table costs the same whether it is one-way or two-way, but there are far more two-way tables, so each generator must choose. The first, which we call **Independent**, measures the 13 one-way tables with equal shares of the budget, and nothing else. Its rows have the right share of each age band and each occupation, but every attribute varies independently of the others, income included. It shows what the other two generators buy.

The second is **MST**, which won the 2018 synthetic data competition of the US National Institute of Standards and Technology (McKenna, Miklau & Sheldon, 2021). It splits the budget in three.

One third of MST's budget measures the one-way tables. One third chooses twelve two-way tables. For that, MST fits a model to the one-way tables and scores each pair of attributes by how badly that model gets their two-way table wrong. It then picks twelve pairs one at a time with the exponential mechanism, so that they form a tree over the 13 attributes. The last third measures

those twelve tables. A tree is the largest set of pairs that Private-PGM can always fit cheaply, so MST stops there.

```
def mst(data: Dataset, rho: float, rng: np.random.Generator) -> Dataset:
    """MST: one-way tables, then a spanning tree of the two-way tables that a one-way
     ↪  model gets most wrong."""
    d = len(data.domain)  # a third of the budget each: the one-way tables, the choice
 ↪  of tree, the tree's tables
    one_way = [measure(data, (a,), math.sqrt(3 * d / (2 * rho)), rng) for a in
 ↪  data.domain]
    model = fit(data.domain, one_way, iters=2500)
    scores = {
        pair: float(np.abs(data.project(pair).datavector() -
 ↪  model.project(pair).datavector()).sum())
        for pair in pairs(data.domain)
    }
    tree: list[Clique] = []
    component = {a: a for a in data.domain}  # which tree component each attribute
 ↪  belongs to
    eps_round = math.sqrt(8 * (rho / 3) / (d - 1))
    for _ in range(d - 1):
        joining = {p: s for p, s in scores.items() if component[p[0]] !=
 ↪  component[p[1]]}
        a, b = exponential_mechanism(joining, eps_round, rng)
        tree.append((a, b))
        component = {x: component[a] if c == component[b] else c for x, c in
 ↪  component.items()}
    two_way = [measure(data, edge, math.sqrt(3 * (d - 1) / (2 * rho)), rng) for edge
 ↪  in tree]
    return fit(data.domain, one_way + two_way, iters=5000).synthetic_data()
```

The third is **AIM**, an adaptive and iterative mechanism that does best on the public benchmarks (McKenna, Mullins, Sheldon & Miklau, 2022). It is told which tables matter, here all 78 two-way tables, and it spends the budget in rounds, 16 per attribute. Nine tenths of each round's share go to measuring and one tenth to choosing. In each round, AIM scores every candidate table by how far its model's answer is from the real one, minus the error that the noise alone would cause. The candidates are the one-way and two-way tables, weighted by how many of the 78 tables they inform, and only tables that keep the model small enough to fit are allowed.

AIM then picks a table with the exponential mechanism, measures it and refits the model. If the measurement did not improve the model's answer by more than the noise explains, the noise is too large to be useful. So AIM halves σ for the remaining rounds, each of which then costs four times more, and the run ends sooner. When the budget cannot cover another full round, the last round spends what is left.

```
measurements = [measure(data, (a,), sigma, rng) for a in dom]
spent = len(dom) * 0.5 / sigma**2
model = fit(dom, measurements, iters=1000)
last = False
while not last:
    if rho - spent < 2 * (0.5 / sigma**2 + eps**2 / 8):  # the next round would
     ↪  overspend: make it the last
```

```
        sigma, eps, last = math.sqrt(1 / (2 * 0.9 * (rho - spent))), math.sqrt(8 * 0.1
↪  * (rho - spent)), True
    spent += eps**2 / 8 + 0.5 / sigma**2
    allowed = allowed_cliques(candidates, model, max_model_size * spent / rho)
    tables = model_tables(model, allowed)  # the model's answer to every candidate,
↪  without the library's compiler
    scores = {c: candidates[c] * (l1(answers[c], tables[c]) - bias(sigma, dom, c)) for
↪  c in allowed}
    clique = exponential_mechanism(scores, eps, rng)
    measurements.append(measure(data, clique, sigma, rng))
    before = error(answers[clique], model, clique)
    measured = list(dict.fromkeys(m.clique for m in measurements))  # a table can be
↪  measured twice
    model = fit(dom, measurements, iters=1000, warm=model.potentials.expand(measured))
    if before - error(answers[clique], model, clique) <= bias(sigma, dom, clique):  #
     ↪  too noisy to help
        sigma, eps = sigma / 2, eps * 2
```

The generators' settings are those of the reference implementations, such as AIM's limit of 80 megabytes on the model. We made one change to MST. The reference MST first merges the rare values of each attribute, such as countries with a handful of customers, but we keep the codebook's lists for all three generators, so that they are compared on the same table. The script also computes AIM's model answers itself, in plain numpy, which is faster than the library and gives the same tables.

The agency trains a logistic regression on whatever table it gets, with the library's default settings, on the 12 attributes other than income as 0-or-1 columns. The same model trained on the real table is the campaign's reference point. For a fair comparison of the two routes, the script also trains DP-SGD on these 12 attributes, with the settings of Section 5.1.

The experiment runs each generator at ten values of ε from 0.03 to 10, with five random seeds each. Each run is one job, made of a technique, a value of ε, a number of customer records and a seed. The script lists the jobs, the slow ones first, and spreads them over the processor cores:

```
jobs: list[tuple[str, float | None, int, int, Path | None]] = [("Real table", None,
↪  TRAIN_ROWS, 0, cache)]
for technique in [*GENERATORS, "DP-SGD"]:
    jobs += [(technique, eps, TRAIN_ROWS, seed, cache) for eps in epsilons for seed in
↪  range(seeds)]
for n in SIZES[:-1]:
    jobs += [("Real table", None, n, 0, cache)]
    jobs += [("AIM", eps, n, seed, cache) for eps in epsilons for seed in
↪  range(size_seeds)]
jobs.sort(key=lambda j: (j[0] != "AIM", -(j[1] or 0.0)))  # the slow jobs first, so the
↪  cores stay busy
context = multiprocessing.get_context("spawn")  # jax must start fresh in each worker
# A worker's memory grows with every program jax compiles, so each job gets a fresh
↪  process.
pool = ProcessPoolExecutor(WORKERS, mp_context=context, initializer=_init,
↪  initargs=(data,), max_tasks_per_child=1)
with pool:
    runs = list(pool.map(evaluate, jobs))
```

6.3 Read the results, technical first

Figure 4 shows the engineer's view, which is how the literature on synthetic data judges a generator. Its vertical axis is the average, over the 78 two-way tables, of the *total variation distance* between the synthetic and the real proportions. That distance is half the sum of the absolute differences of the proportions. It is the largest error the synthetic table can make on any question about that pair of attributes, as a share of the customers.

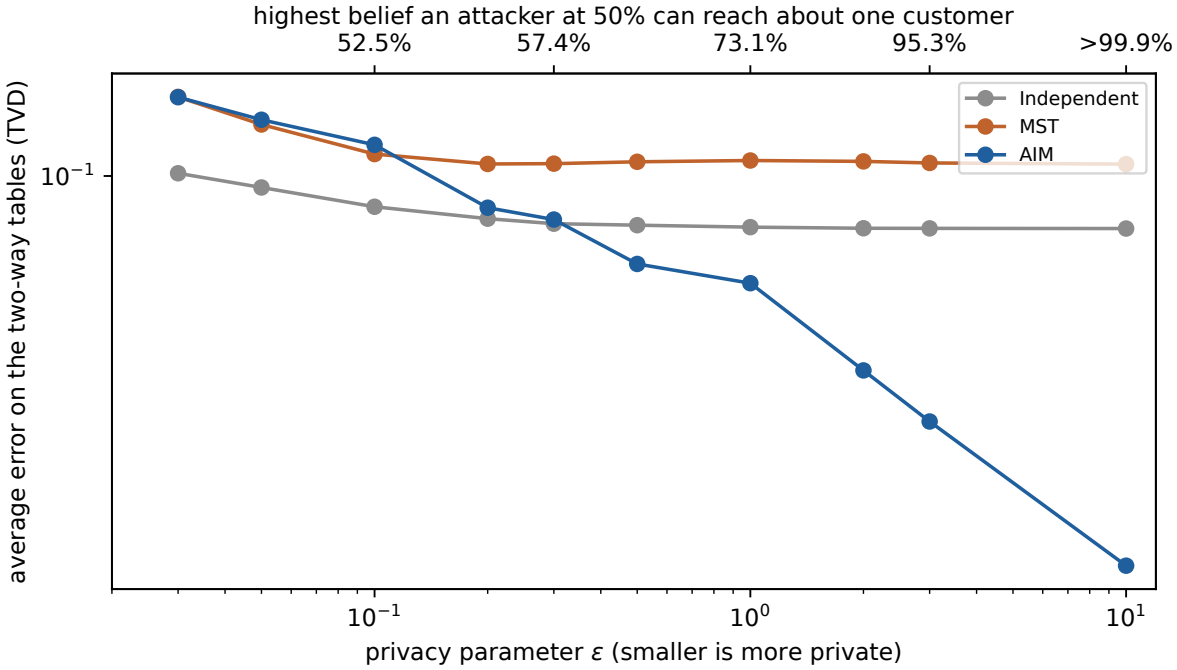


Figure 4: Error on the two-way tables against the privacy setting. Each line is a generator, averaged over five seeds; the vertical axis is logarithmic.

AIM's error falls steadily as the budget grows, from 0.151 at $\varepsilon = 0.03$ to 0.057 at $\varepsilon = 1$ and 0.013 at $\varepsilon = 10$, because a larger budget buys more rounds and less noise in each. MST's error stops falling at about 0.11 from $\varepsilon = 0.2$ on. It measures only the 12 two-way tables of its tree, so the other 66 are guesses from the model, and no budget improves a guess. Independent measures no two-way table, and its error settles at 0.076, the error of treating every pair of attributes as unrelated. Up to $\varepsilon = 0.3$ that guess beats both other generators, which spread too little budget over too many tables. From $\varepsilon = 0.5$ on, AIM has the smallest error.

Figure 5 shows the first business count, the eligible customers that the synthetic plan puts in the wrong segment, out of 7,841. The shaded bands run from the 10th to the 90th percentile over the seeds. The plan is a three-way table, age band by occupation by income, and no generator measures it directly.

The plan tells a different story from the average error. Independent misplaces about 3,000 of the 7,841 at every budget, because it treats income as unrelated to age and occupation. MST and AIM do worse than that below $\varepsilon = 1$, misplacing between 3,290 and 5,690. Their models link income to the segment attributes, but through noisy or indirect tables, which can be worse than no link.

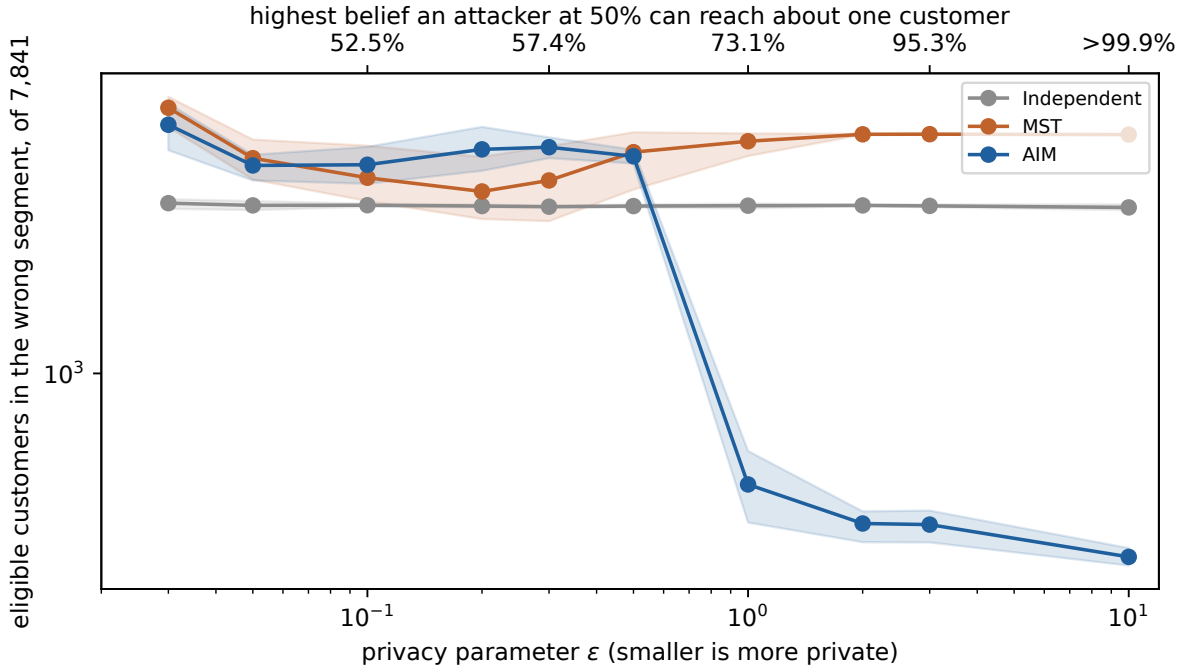


Figure 5: Eligible customers placed in the wrong segment of the campaign plan, out of 7,841, against the privacy setting. The vertical axis is logarithmic.

At $\varepsilon = 1$ AIM's error drops to 484 misplaced customers, 6.2% of them, and it stays below 400 from $\varepsilon = 2$ on. With that budget AIM can afford the tables that tie income to age and to occupation. So a synthetic plan is usable only from about $\varepsilon = 1$, and only with AIM.

Figure 6 shows the campaign count. The upper dashed line is the agency's model trained on the real table, and the dotted line is no data at all. The dashed line with square markers is the model route, DP-SGD trained inside the bank on the same 12 attributes at the same ε.

AIM's table gives the best campaign at most budgets. It reaches about 1,900 to 2,240 of the 3,846 eligible customers up to $\varepsilon = 0.1$, about 1,830 to 2,030 between 0.2 and 0.5, and 2,245 at $\varepsilon = 1$, rising to 2,513 at $\varepsilon = 10$. In the middle AIM does worse than at $\varepsilon = 0.1$, beyond the spread over seeds, and at $\varepsilon = 0.2$ MST beats it, 2,081 to 1,828. A larger budget makes AIM measure more tables, and the extra tables do not always help the campaign. Independent stays near random offers, between 631 and 1,125, because a table without links between attributes gives a model nothing to learn. The DP-SGD model beats every synthetic table at every budget. It reaches 2,620 at $\varepsilon = 1$, within 5 of the model trained on the real table, while AIM's table at $\varepsilon = 1$ is 380 short.

Table 6 lists every synthetic configuration, with the frontier defined as in Section 5.2.

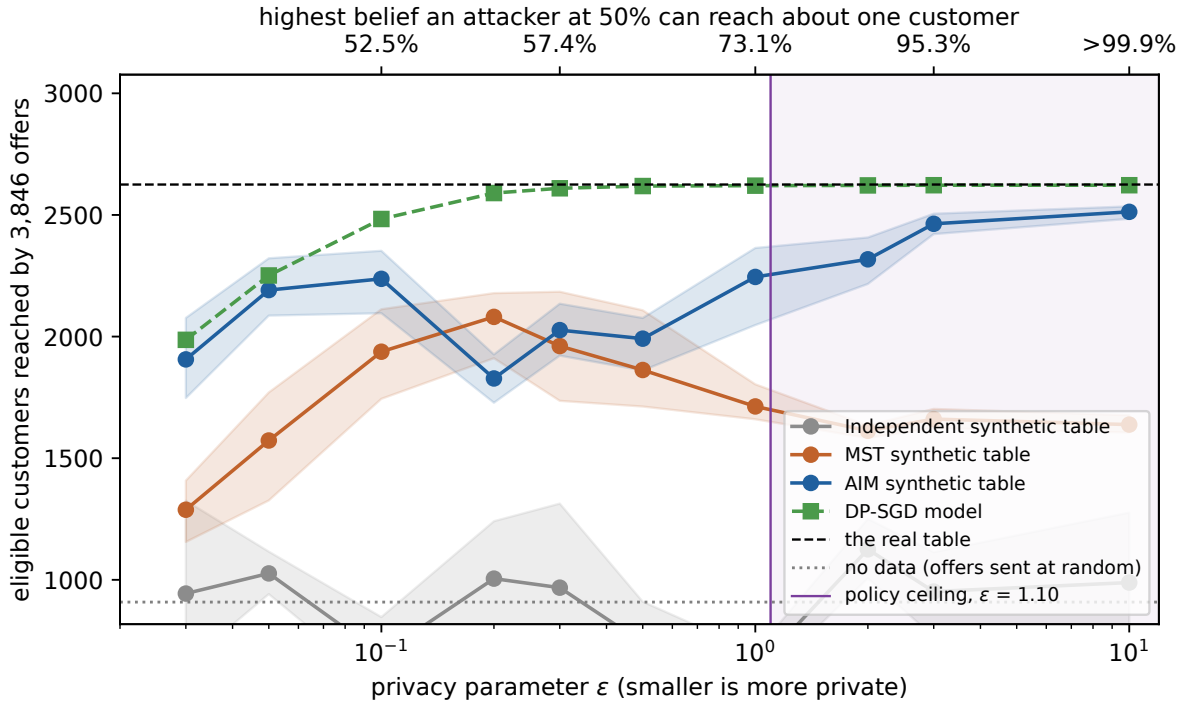


Figure 6: Eligible customers reached by the campaign against the privacy setting, for a model trained on each synthetic table and for a DP-SGD model trained on the records directly. The shaded region is ruled out by the policy ceiling.

Table 6: Every synthetic configuration, averaged over the seeds.

Generator	ε	Attacker's belief ceiling	Two-way error	Eligible misplaced in the plan	Eligible reached	On the frontier
Independent	0.03	50.7%	0.101	3,048 of 7,841	944 of 3,846 (25%)	
MST	0.03	50.7%	0.151	5,690 of 7,841	1,288 of 3,846 (33%)	
AIM	0.03	50.7%	0.151	5,092 of 7,841	1,906 of 3,846 (50%)	yes
Independent	0.05	51.2%	0.094	3,004 of 7,841	1,026 of 3,846 (27%)	
MST	0.05	51.2%	0.131	4,091 of 7,841	1,573 of 3,846 (41%)	
AIM	0.05	51.2%	0.134	3,901 of 7,841	2,191 of 3,846 (57%)	yes
Independent	0.1	52.5%	0.085	3,007 of 7,841	694 of 3,846 (18%)	
MST	0.1	52.5%	0.112	3,599 of 7,841	1,938 of 3,846 (50%)	
AIM	0.1	52.5%	0.118	3,919 of 7,841	2,237 of 3,846 (58%)	yes
Independent	0.2	55.0%	0.080	2,991 of 7,841	1,005 of 3,846 (26%)	
MST	0.2	55.0%	0.106	3,290 of 7,841	2,081 of 3,846 (54%)	
AIM	0.2	55.0%	0.085	4,334 of 7,841	1,828 of 3,846 (48%)	
Independent	0.3	57.4%	0.078	2,978 of 7,841	968 of 3,846 (25%)	
MST	0.3	57.4%	0.107	3,536 of 7,841	1,961 of 3,846 (51%)	
AIM	0.3	57.4%	0.080	4,396 of 7,841	2,027 of 3,846 (53%)	

Generator	ε	Attacker's belief ceiling	Two-way error	Eligible misplaced in the plan	Eligible reached	On the frontier
Independent	0.5	62.2%	0.077	2,990 of 7,841	741 of 3,846 (19%)	
MST	0.5	62.2%	0.108	4,255 of 7,841	1,863 of 3,846 (48%)	
AIM	0.5	62.2%	0.063	4,141 of 7,841	1,991 of 3,846 (52%)	
Independent	1	73.1%	0.077	2,998 of 7,841	631 of 3,846 (16%)	
MST	1	73.1%	0.108	4,570 of 7,841	1,713 of 3,846 (45%)	
AIM	1	73.1%	0.057	484 of 7,841	2,245 of 3,846 (58%)	yes
Independent	2	88.1%	0.076	3,004 of 7,841	1,125 of 3,846 (29%)	
MST	2	88.1%	0.108	4,783 of 7,841	1,613 of 3,846 (42%)	
AIM	2	88.1%	0.036	374 of 7,841	2,318 of 3,846 (60%)	yes
Independent	3	95.3%	0.076	2,994 of 7,841	952 of 3,846 (25%)	
MST	3	95.3%	0.107	4,786 of 7,841	1,661 of 3,846 (43%)	
AIM	3	95.3%	0.028	372 of 7,841	2,464 of 3,846 (64%)	yes
Independent	10	100.0%	0.076	2,964 of 7,841	989 of 3,846 (26%)	
MST	10	100.0%	0.106	4,775 of 7,841	1,638 of 3,846 (43%)	
AIM	10	100.0%	0.013	301 of 7,841	2,513 of 3,846 (65%)	yes

6.4 Check how much the data size matters

A measured table carries noise of a fixed size whatever the number of customers, so with fewer customers the same noise is a larger share of every count. To measure the effect, the script runs AIM again on the first 2,000 and the first 8,000 customer records, and Figure 7 compares the three sizes on the campaign count.

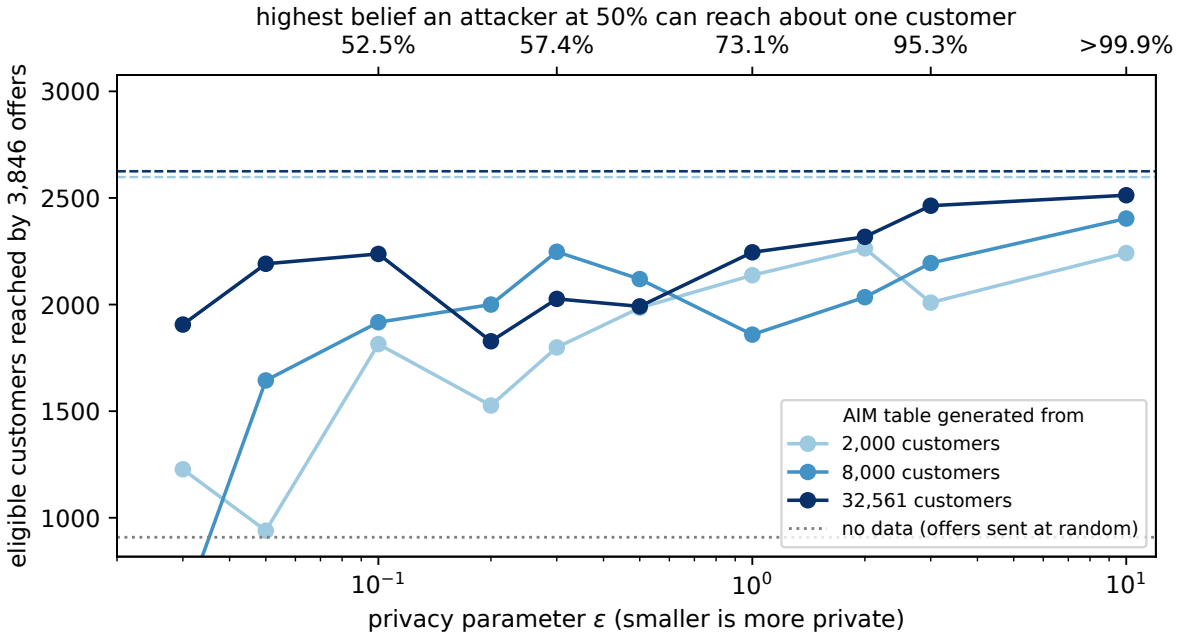


Figure 7: Eligible customers reached by the campaign with an AIM table generated from 2,000, 8,000 and 32,561 customer records. Each dashed line is the agency's model trained on the real records of that size.

Fewer records reduce campaign performance most at the smallest budget. At $\varepsilon = 0.1$, AIM's table misses 784 eligible customers with 2,000 training records, 709 with 8,000 and 388 with all 32,561, as Table 7 shows. Results at larger budgets are less certain because the smaller datasets used only two seeds each. For example, the run with 8,000 records misses 767 customers at $\varepsilon = 1$, compared with 460 for 2,000 records; this reversal may reflect variation between runs. A bank with only a few thousand customers should run the experiment at its own data size before committing to a guarantee.

Table 7: Eligible customers missed because of privacy with an AIM table, by the number of customer records it is generated from.

Customers in the records	Reached with the real table	Missed at $\varepsilon = 0.1$	Missed at $\varepsilon = 0.3$	Missed at $\varepsilon = 1$
2,000	2,598	784	799	460
8,000	2,626	709	378	767
32,561	2,625	388	598	380

7 Step 6: compare the routes and decide

The decision maker needs a few options, each with a privacy sentence and its business effect, and Table 8 gives them. Each option is AIM's table at a privacy level, next to the DP-SGD model at the same level, and the two reference points are no data and the real table. Here DP-SGD is trained on the synthetic route's 12 banded attributes, not on the features of Section 5.1, so its rows differ a little from Section 5.2, for example 2,620 instead of 2,552 at $\varepsilon = 1$. The membership column gives a second privacy sentence. It bounds how often an attack can correctly identify a real customer, for any attack that wrongly accuses 1% of non-customers (Kairouz et al., 2015).

Table 8: The options menu for one campaign of 3,846 offers: the best generator at each option, and the model route at the same privacy level.

Option	Attacker's belief ceiling	Member-ship TPR bound at 1% FPR	Eligible misplaced in the plan	Eligible reached	Missed vs. the real table	Gained vs. no data
No data (random offers)	—	—	—	909	1,716	0
$\varepsilon = 0.1$, AIM table	52.5%	1.1%	3,919 of 7,841	2,237	388	1,329
$\varepsilon = 0.1$, DP-SGD model	52.5%	1.1%	no plan	2,483	142	1,575
$\varepsilon = 0.3$, AIM table	57.4%	1.4%	4,396 of 7,841	2,027	598	1,118
$\varepsilon = 0.3$, DP-SGD model	57.4%	1.4%	no plan	2,610	15	1,701
$\varepsilon = 1$, AIM table	73.1%	2.7%	484 of 7,841	2,245	380	1,337
$\varepsilon = 1$, DP-SGD model	73.1%	2.7%	no plan	2,620	5	1,712
$\varepsilon = 3$, AIM table	95.3%	20.1%	372 of 7,841	2,464	161	1,555
$\varepsilon = 3$, DP-SGD model	95.3%	20.1%	no plan	2,622	3	1,714
The real table, no DP	no limit	no limit	0	2,625	0	1,716

The policy ceiling from Section 3, $\varepsilon \leq 1.10$, picks the option on each route. On the model route it gives this result:

The largest value on the grid below the ceiling is $\varepsilon = 1$, and at that value the DP-SGD model is on the frontier. Across the seeds it reaches between 2,544 and 2,557 of the 3,846 eligible customers, 2,552 on average. The model without DP reaches 2,568, and random offers reach 909. So privacy costs about 16 eligible customers per campaign, while the private model still reaches about 1,643 more than sending the offers at random. In money, with V the value to the bank of reaching one eligible customer, the cost of privacy is about $16\,V$ per campaign and the value of the private model over no model is about $1,643\,V$. Even at $\varepsilon = 10$, where the noise is small, the private model reaches 2,552. So almost all of the cost at the ceiling comes from clipping each customer's influence, which DP-SGD does at every ε, and almost none from the noise that the stricter policy adds.

On the synthetic-data route it gives this result:

The largest value on the grid below the ceiling is $\varepsilon = 1$, and at that value the AIM table is on the frontier. Across the seeds its campaign reaches between 2,047 and 2,364 of the 3,846 eligible customers, 2,245 on average, and its plan puts 484 of the 7,841 eligible customers in the records into the wrong segment, 6.2% of them. The real table reaches 2,625 and random offers reach 909. So privacy costs about 380 eligible customers per campaign, while the synthetic table is still worth about 1,337 customers more than no data. In money, with V the value to the bank of reaching one eligible customer, the cost of privacy is about $380\,V$ per campaign and the value of the table over no data about $1,337\,V$. The model route, a DP-SGD model on the same attributes at the same ε, reaches 2,620, so the table costs about 375 more customers per campaign than the model, and in return it answers the planning question and any other the agency asks.

If the agency needs only the targeting model, the model route wins outright, because at the same guarantee it reaches 2,620 eligible customers to the table's 2,245. The table is worth its extra cost only because it also answers the planning question, and at $\varepsilon = 1$ it answers it well, with 6.2% of the eligible customers misplaced. Below $\varepsilon = 1$ the synthetic plan misplaces more than 40% of them, so the table is then hard to justify over the model. So the bank releases the AIM table at $\varepsilon = 1$ if the agency's plan matters to it, and the DP-SGD model if it does not. In money, the table costs about $375\,V$ more per campaign than the model, and the bank compares that with the value of the plan.

Several people sign the decision, each for their own part. The privacy officer signs the ceiling, fixed before the results were known. The engineer signs that the release meets the ceiling: for a model, with the accountant's ε and δ, and for a table, with the budget ρ and its conversion. The business owner signs the cost in customers, and so in money, and on the synthetic route the agency signs the share of the plan it accepts. A release record keeps all of this with the scripts, so that anyone can check the decision and revisit it when the data or the policy changes.

8 Doing this in another business

The six steps carry over to any use case, and only their contents change. The counts are whatever the decision turns on, such as fraud caught at a fixed number of reviews, or the segment sizes and forecasts that a recipient will compute from a table. The baseline is what the business does without the release. The ceiling is still a sentence about an attacker's belief, fixed before anyone sees the results, and a generator's budget follows from it. The output is still a short menu with a

cost in the business's own units, which puts a model and a table side by side for the people who own the risk and the revenue to sign.

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